Intercarrier Interference in OFDM: a Deterministic Model for Transmissions in Mobile Environments with Imperfect Synchronization

Christian Oberli [1], Martín García [2]

Department of Electrical Engineering, Pontificia Universidad Católica de Chile
Vicuña Mackenna 4860, Santiago, Chile
[1] obe@ing.puc.cl
[2] mgarcia@ing.puc.cl

***Abstract*—Intercarrier Interference (ICI) is an impairment well known to degrade performance of Orthogonal Frequency Division Multiplexing (OFDM) transmissions. It arises from carrier frequency offsets (CFO), from the Doppler spread due to channel time-variation, and, to a lesser extent, from sampling frequency offsets (SFO).**

The literature reports several models of ICI due to each kind of impairment. Some studies describe ICI due to two of the three impairments, but so far no general model exists to describe the joint effect of all three impairments together. Furthermore, most available models involve some level of approximation, and the diversity of approaches makes it cumbersome to compare power levels of the different kinds of ICI.

In this work, we present the derivation of a general and mathematically exact model for the ICI stemming from the joint effect of the three impairments mentioned.

I. Introduction

Mathematical models of Intercarrier Interference in Orthogonal Frequency Division Multiplexing (OFDM) and techniques for mitigating it have been reported by many authors. The case of time-varying channels is addressed in [1]–[9], of CFO in [10]–[15] and of SFO in [16], [17]. Work modeling ICI produced jointly by two of the three impairments is significantly less common. The joint effect of CFO and SFO has been studied in [18], [19], while [20] reports on ICI due to CFO and channel mobility. Despite the substantial attention that modeling and dealing with ICI has received so far, there is as yet no general model in the literature that describes ICI resulting from the joint effect of all three impairments.

Additionally, many of the above cited references (and others cited therein) model ICI by using discrete-domain signal treatments. This approach is perhaps favored because OFDM is, at its very heart, a discrete-domain operation (the discrete Fourier transform). Unfortunately, however, discrete-domain approaches are often inaccurate for representing impairments whose nature is clearly continuous, such as the ones mentioned.

In this paper we derive a general model of ICI for OFDM transmissions subject to the joint effect of CFO, SFO and channel time-variation. The model we find is deterministic for a given time-varying channel. It is also mathematically exact, and is therefore devoid of both limitations mentioned above.

II. Deterministic model of ICI

In what follows, we derive a signal model for OFDM transmissions that include the effects of CFO, SFO and channel mobility.

We begin by modeling the signal of the mth OFDM symbol in continuous time t using complex baseband notation as

$$x(t) = \sqcap\left(\frac{t-\left(m+\frac{1}{2}\right)T_g+T_e}{T_g}\right) \cdot \sum_{\nu=-\frac{N_s}{2}}^{\frac{N_s}{2}-1} X_m(\nu)e^{\jmath 2\pi\nu(t-mT_g)\delta_t}. \tag{1}$$

In this equation, $X_m(\nu)$ is the modulation on subcarrier ν of a total of N_s subcarriers. The separation between subcarriers is $\delta_t = \frac{1}{N_sT}$ Hertz, where T is the sampling period of the transmitter. The cyclic prefix has N_e samples and a duration of $T_e = N_eT$ seconds. Thus, the complete OFDM symbol has $N_g = N_e + N_s$ samples and a duration of $T_g = N_gT$ seconds. The symbol $\jmath$ denotes the imaginary unit $\sqrt{-1}$. Finally, $\sqcap(x)$ is the rectangular function, equal to 1 when x is between $\left[-\frac{1}{2}, \frac{1}{2}\right]$ and 0 elsewhere.

In [21] and [22], the input-output relationship of a time-variant linear system is described as

$$y(t) = \int_0^\infty h(t,\tau)x(t-\tau)d\tau, \tag{2}$$

where $x(t)$ and $y(t)$ are the respective input and output signals in the time domain. The function $h(t,\tau)$ is the time-variant impulse response of the system observed at instant t due to an impulse at time $t-\tau$.

Now consider $h(t,\tau)$ to represent a time-varying wireless channel, and substitute (1) for $x(t)$ in (2). Then, following the steps outlined in Appendix A for including the effects of a CFO of Δf Hertz and an SFO of ΔT in the received signal, we obtain

$$
\begin{aligned}
y(t) &= \\
&(T+\Delta T)\sum_{p=0}^{N_s-1}\sum_{\nu=-\frac{N_s}{2}}^{\frac{N_s}{2}-1} X_m(\nu)e^{\jmath 2\pi\Delta f t}e^{-\jmath 2\pi\nu\delta_t m T_g}\\
&\cdot\,\delta\left(t-p(T+\Delta T)-mN_g(T+\Delta T)\right)\\
&\cdot\left[\int_0^\infty h(t,\tau)e^{\jmath 2\pi\nu\delta_t(t-\tau)}\right.\\
&\left.\cdot\sqcap\left(\frac{t-\left(\frac{1}{2}N_s+mN_g\right)(T+\Delta T)}{N_s(T+\Delta T)}\right)d\tau\right]\\
&+\text{Ш}\left(\frac{t}{T+\Delta T}\right)\cdot n(t). \qquad (3)
\end{aligned}
$$

Above, $\delta(\cdot)$ is the impulse function, $\text{Ш}(\cdot)$ is the sampling function and $n(t)$ is Additive White Gaussian Noise (AWGN). Equation (3) is a continuous-time signal. It is continuous because so is the function $\text{Ш}(\cdot)$, but its value is 0 at every instant except when $t=p(T+\Delta T)+mN_g(T+\Delta T)$, with $p=0,\ldots,N_s-1$.

To recover the symbols in an actual OFDM system implementation, the Fast Fourier Transform (FFT) of the N_s samples is calculated. That operation is equivalent to calculating the continuous Fourier transform of (3) with respect to t, but with the origin fixed at time $mN_g(T+\Delta T)$ (i.e. taking the transform of $y(t+mN_g(T+\Delta T))$). Upon following the algebraic steps detailed in Appendix B, we obtain

$$
\begin{aligned}
Y(f) \;=\; &\left(\int_0^\infty\left[s^{\sqcap}(f,\tau)*_f\left\{(T+\Delta T)\sum_{\nu=-\frac{N_s}{2}}^{\frac{N_s}{2}-1}X_m(\nu)\right.\right.\right.\\
&\cdot e^{-\jmath 2\pi(f-\Delta f-\nu\delta_t)(T+\Delta T)\left(mN_g+\frac{N_s-1}{2}\right)}\\
&\cdot\frac{\sin[\pi(f-\Delta f-\nu\delta_t)(T+\Delta T)N_s]}{\sin[\pi(f-\Delta f-\nu\delta_t)(T+\Delta T)]}\\
&\left.\left.\left.\cdot e^{-\jmath 2\pi\nu\delta_t\tau}e^{-\jmath 2\pi\nu\delta_t m T_g}\right\}\right]d\tau\right)e^{\jmath 2\pi f m N_g(T+\Delta T)}\\
&+N(f). \qquad (4)
\end{aligned}
$$

In this equation the function $s^{\sqcap}(f,\tau)$ is the Doppler-variant impulse response [21], time-limited to the duration of one OFDM symbol. Thus, $s^{\sqcap}(f,\tau)$ is given by

$$
\begin{aligned}
s^{\sqcap}(f,\tau) &= \int_{mN_g(T+\Delta T)}^{(N_s+mN_g)(T+\Delta T)} h(t,\tau)e^{-\jmath 2\pi f t}dt\\
&= N_s(T+\Delta T)\,\mathrm{sinc}\left(fN_s(T+\Delta T)\right)\\
&\quad\cdot e^{-\jmath 2\pi f\left(\frac{1}{2}N_s+mN_g\right)(T+\Delta T)} *_f s(f,\tau), \quad (5)
\end{aligned}
$$

where $s(f,\tau)$ is the Doppler-variant impulse response defined as the Fourier transform in t of $h(t,\tau)$ [21]. For a fixed delay τ, and if the channel is static, then $s(f,\tau)$ is a frequency domain impulse. As channel mobility increases, so does the frequency spread of $s(f,\tau)$. The symbol $*_f$ denotes continuous convolution in the frequency domain.

Expression (4) gives an exact description of the continuous spectrum of an OFDM signal received over a time-variant channel with a CFO of Δf Hertz and an SFO of ΔT seconds. In practice, this signal is observed at the output of the FFT in the receiver at frequencies of $f=l\delta_r$, with $-\frac{N_s}{2}\le l\le\frac{N_s}{2}-1$ and $\delta_r=\frac{1}{N_s(T+\Delta T)}$ equal to the separation of the subcarrier frequencies used by the receiver. Imposing these conditions on (4) and considering an arbitrary subcarrier k, we obtain (Appendix C) the discrete output of the system:

$$
\begin{aligned}
Y(k) &= \lambda(k,k)\bar{H}(k)X_m(k)+I(k)\\
&\quad+W(k)+Q(k)+N(k), \qquad (6)
\end{aligned}
$$

where $\lambda(\cdot,\cdot)$ is a phase and magnitude distortion given by

$$
\begin{aligned}
\lambda(d,\nu) &= \frac{1}{N_s}e^{\jmath 2\pi(d-\nu)\delta_t m T_b}\\
&\cdot e^{-\jmath 2\pi\left[\frac{d-\nu}{N_s}-\frac{\nu}{N_s}\frac{\Delta T}{T}-\Delta f T\left(1+\frac{\Delta T}{T}\right)\right]\left(mN_b+\frac{N_s-1}{2}\right)}\\
&\cdot\frac{\sin\left\{\pi\left[\frac{d-\nu}{N_s}-\frac{\nu}{N_s}\frac{\Delta T}{T}-\Delta f T\left(1+\frac{\Delta T}{T}\right)\right]N_s\right\}}{\sin\left\{\pi\left[\frac{d-\nu}{N_s}-\frac{\nu}{N_s}\frac{\Delta T}{T}-\Delta f T\left(1+\frac{\Delta T}{T}\right)\right]\right\}}, \quad (7)
\end{aligned}
$$

and $\bar{H}(k)$ in (6) is the time domain average of the channel in carrier k during the transmission of the m-th OFDM symbol, given by

$$
\begin{aligned}
\bar{H}(k) &= \frac{1}{N_s(T+\Delta T)}\cdot\int_0^\infty s^{\sqcap}(0,\tau)e^{-\jmath 2\pi k\delta_t\tau}d\tau \qquad (8)\\
&= \int_0^\infty\frac{e^{-\jmath 2\pi k\delta_t\tau}}{N_s(T+\Delta T)}\int_{mN_g(T+\Delta T)}^{(N_s+mN_g)(T+\Delta T)} h(t,\tau)\;dt\;d\tau.
\end{aligned}
$$

Finally $I(k)$, $W(k)$ and $Q(k)$ in (6) represent various forms of inter-carrier interference (ICI). Concretely, $I(k)$ is ICI solely due to mobility (equals zero if the channel is static), $W(k)$ is ICI caused exclusively by imperfect synchronization (equals zero if there are no frequency nor sampling offsets), and $Q(k)$ is an ICI that is non-zero only when both impairments are present. Their expressions are

$$
\begin{aligned}
I(k) &= \frac{1}{N_s(T+\Delta T)}\sum_{\substack{d=-\frac{N_s}{2}\\ d\neq k}}^{\frac{N_s}{2}-1}X_m(d)\lambda(d,d)\\
&\quad\cdot\int_0^\infty s^{\sqcap}((k-d)\delta_r,\tau)e^{-\jmath 2\pi d\delta_t\tau}d\tau, \qquad (9)
\end{aligned}
$$

$$
W(k)=\sum_{\substack{\nu=-\frac{N_s}{2}\\ \nu\neq k}}^{\frac{N_s}{2}-1}X_m(\nu)\lambda(k,\nu)\bar{H}(\nu), \qquad (10)
$$

and

$$
\begin{aligned}
Q(k) &= \frac{1}{N_s(T+\Delta T)}\sum_{\substack{d=-\frac{N_s}{2}\\ d\neq k}}^{\frac{N_s}{2}-1}\sum_{\substack{\nu=-\frac{N_s}{2}\\ \nu\neq d}}^{\frac{N_s}{2}-1}X_m(\nu)\lambda(d,\nu)\\
&\quad\cdot\int_0^\infty s^{\sqcap}((k-d)\delta_r,\tau)e^{-\jmath 2\pi\nu\delta_t\tau}d\tau. \qquad (11)
\end{aligned}
$$

Results (6) through (11) are deterministic and mathematically exact. In (9) (ICI due to mobility) we observe that the interference in subcarrier k is the sum of signals from all the other subcarriers, respectively weighted by the integral of $s^{\sqcap}((k-d)\delta_r, \tau)$. The value of the integral depends on mobility and on the separation between the interfering subcarriers (d) and the desired subcarrier (k). This value is relevant only when subcarrier d is in the neighborhood of k. The size of the neighborhood grows with the mobile's speed, but in any current-day OFDM systems designed for mobility (e.g. DVB-T/H [23], "mobile WiMAX" [24]), the neighborhood is mainly comprised by subcarriers $k+1$ and $k-1$. If synchronization is perfect (that is, $\Delta T = 0$ and $\Delta f = 0$), then (9) is similar to the description found by many authors [2], [6], [9], [25] for representing interference based on Doppler-variant impulse responses. However, they all use discrete-domain approaches, different from the one employed here, thus capturing the effect of ICI less accurately.

Term (10) (ICI due to imperfect synchronization) has been described by several authors for static channel conditions, either considering CFO and SFO jointly [18], [19], CFO alone [10], [12], or SFO alone [16], [17].

Finally, $Q(k)$ of (11) is a new finding, and clarifies a frequent misconception that ICI due to mobility and imperfect synchronization are two separate additive terms. There is an ICI enhancement when both impairments are present. Inspection of the above equations, however, supports the intuition that $Q(k)$ is an ICI component significantly weaker than $I(k)$ and $W(k)$ —a precise quantification will be addressed in future work, which will focus on studying the statistical behavior of (9), (10) and (11), and on comparing the power levels of these three types of ICI under typical operating conditions of OFDM systems in mobile environments with imperfect synchronization.

III. Conclusions

A general and mathematically exact model of the power of inter-carrier interference (ICI) was derived for OFDM transmissions exposed to the joint impact of sampling frequency offset, carrier frequency offset (CFO) and channel time variation. It was shown that the ICI ensuing from these impairments has three components: one solely caused by Doppler spread, one that depends only on the synchronization offsets, and one that is non-zero only when imperfect synchronization and channel variation happen together. Similar but nevertheless approximate descriptions of the former two components are available in the literature. In this paper, besides describing them without approximations, they are presented with the same power scale. This allows for a direct comparison of ICI caused by different impairments. The third component is a new finding that corrects the commonly accepted view that ICI due to channel time-variation and imperfect synchronization are entirely separable. This component, however, is expected to be significantly less powerful than the other two.

Appendix A

By substituting (1) into (2) and adding the phasor $e^{\jmath 2\pi \Delta f t}$ due to the difference of Δf Hertz between the receiver and transmitter carrier frequencies, we obtain

$$y(t) = e^{\jmath 2\pi\Delta f t} \int_0^\infty \sqcap\left(\frac{t-\tau-(m+\frac{1}{2})T_g+T_e}{T_g}\right) \cdot \sum_{\nu=-\frac{N_s}{2}}^{\frac{N_s}{2}-1} X_m(\nu) e^{\jmath 2\pi\nu\delta_t(t-\tau-mT_g)} h(t,\tau) d\tau + n(t). \tag{12}$$

We now sample the received signal (12) at a rate offset by ΔT seconds from the transmitter rate, that is, at instants $n\cdot(T+\Delta T)$, where T is the transmitter sampling period. If we assume adequate filtering of the signal was conducted prior to sampling so that the noise term $n(t)$ is limited to the band of interest, the sampling times for the mth OFDM symbol are $t = n(T+\Delta T) + mN_g(T+\Delta T)$, where $n = 0, \ldots, N_s - 1$. Thus, the sampling operation implicitly removes the cyclic prefix and extracts the OFDM symbol in the "correct" window except for a cumulative drift due to ΔT. The sampling function $\sqcup\!\sqcup(\cdot)$ representing this operation is given by

$$\sqcup\!\sqcup\left(\frac{t}{T+\Delta T}\right) = (T+\Delta T) \sqcap\left(\frac{t-\left(\frac{1}{2}N_s+mN_g\right)(T+\Delta T)}{N_s(T+\Delta T)}\right) \cdot \sum_{p=-\infty}^{\infty} \delta\left(t-p(T+\Delta T)-mN_g(T+\Delta T)\right), \tag{13}$$

where $\delta(\cdot)$ is the Dirac delta function. The scaling factor $T+\Delta T$ indicates sampling by area. Thus, applying (13) to (12) we obtain

$$\begin{aligned} y(t) = & \sum_{p=-\infty}^{\infty} \delta\left(t-p(T+\Delta T)-mN_g(T+\Delta T)\right) \\ & \cdot(T+\Delta T) \sum_{\nu=-\frac{N_s}{2}}^{\frac{N_s}{2}-1} X_m(\nu) e^{\jmath 2\pi\Delta f t} \\ & \underbrace{\cdot\sqcap\left(\frac{t-\left(\frac{1}{2}N_s+mN_g\right)(T+\Delta T)}{N_s(T+\Delta T)}\right)}_{\sqcap(\clubsuit)} \\ & \cdot\int_0^\infty \underbrace{\sqcap\left(\frac{t-\tau-\left(m+\frac{1}{2}\right)T_g+T_e}{T_g}\right)}_{\sqcap(\spadesuit)} h(t,\tau) \\ & \cdot e^{\jmath 2\pi\nu\delta_t(t-\tau-mT_g)} d\tau + \sqcup\!\sqcup\left(\frac{t}{T+\Delta T}\right)\cdot n(t), \end{aligned} \tag{14}$$

where $mN_g(T+\Delta T) \le t \le (mN_g+N_s-1)(T+\Delta T)$. If the cyclic prefix duration is sufficiently large that $h(t,\tau) \equiv 0$ for all $\tau > T_e$ at every instant t, a careful analysis of (14) will show that the presence of $\sqcap(\clubsuit)$ allows $\sqcap(\spadesuit)$ to be eliminated. This is so due to the condition on T_e and the position of the $\sqcap(\cdot)$ functions when m and ΔT are within the range of interest. Further simplification may be achieved by joining $\sqcap(\clubsuit)$ with the series in p, as given in (3).

Appendix B

If we calculate the transform of $y(t)$ in (3) evaluated at $t+mN_g(T+\Delta T)$ we obtain

$$Y(f) = \int_0^\infty \underbrace{\int_{-\infty}^{\infty} \sqcap(\clubsuit)h(t,\tau)\psi(t,\tau)e^{-\jmath 2\pi ft}dt\,d\tau}_{\mathcal{F}_t\{\sqcap(\clubsuit)h(t,\tau)\psi(t,\tau)\}} \cdot e^{\jmath 2\pi fmN_g(T+\Delta T)} + N(f), \tag{15}$$

where the function $\psi(t,\tau)$ is defined as

$$\psi(t,\tau) = (T+\Delta T)\sum_{p=0}^{N_s-1}\sum_{\nu=-\frac{N_s}{2}}^{\frac{N_s}{2}-1} X_m(\nu) \cdot\delta\left(t-p(T+\Delta T)-mN_g(T+\Delta T)\right) \cdot e^{\jmath 2\pi\Delta ft}e^{-\jmath 2\pi\nu\delta_t mT_g}e^{\jmath 2\pi\nu\delta_t(t-\tau)}. \tag{16}$$

In (15), $\mathcal{F}_t\{\cdot\}$ is the Fourier transform in t of the product $\sqcap(\clubsuit)h(t,\tau)\psi(t,\tau)$ for a delay of τ in the time-variant impulse response. The transform is given by

$$\mathcal{F}_t\left\{\sqcap(\clubsuit)h(t,\tau)\psi(t,\tau)\right\} = s^{\sqcap}(f,\tau)*_f \Psi(f,\tau), \tag{17}$$

where $\Psi(f,\tau)$ and $s^{\sqcap}(f,\tau)$ are the Fourier transforms in t of $\psi(t,\tau)$ and $\sqcap(\clubsuit)h(t,\tau)$, respectively, given by

$$s^{\sqcap}(f,\tau) = \int_{mN_g(T+\Delta T)}^{(N_s+mN_g)(T+\Delta T)} h(t,\tau)e^{-\jmath 2\pi ft}dt \tag{18}$$

and

$$\Psi(f,\tau) = (T+\Delta T)\sum_{\nu=-\frac{N_s}{2}}^{\frac{N_s}{2}-1} X_m(\nu)e^{-\jmath 2\pi\nu\delta_t mT_g}e^{-\jmath 2\pi\nu\delta_t\tau} \cdot e^{-\jmath 2\pi(f-\Delta f-\nu\delta_t)mN_g(T+\Delta T)} \cdot\sum_{p=0}^{N_s-1} e^{-\jmath 2\pi(f-\Delta f-\nu\delta_t)p(T+\Delta T)}. \tag{19}$$

The last step is to directly evaluate the geometric series in (19), following [19]. We are then left with

$$\Psi(f,\tau) = (T+\Delta T)\sum_{\nu=-\frac{N_s}{2}}^{\frac{N_s}{2}-1} X_m(\nu)e^{-\jmath 2\pi\nu\delta_t mT_g}e^{-\jmath 2\pi\nu\delta_t\tau} \cdot e^{-\jmath 2\pi(f-\Delta f-\nu\delta_t)(T+\Delta T)\left(mN_g+\frac{N_s-1}{2}\right)} \cdot\frac{\sin[\pi(f-\Delta f-\nu\delta_t)(T+\Delta T)N_s]}{\sin[\pi(f-\Delta f-\nu\delta_t)(T+\Delta T)]}. \tag{20}$$

Substituting results (18) and (20) into (15) and (17), we obtain the desired signal model in (4).

Appendix C

To derive the discrete output in (4) for the specified frequencies, we consider, following [19], that

$$e^{-\jmath 2\pi\nu\delta_t mT_g}e^{\jmath 2\pi fmN_g(T+\Delta T)} = e^{\jmath 2\pi(l-\nu)\delta_t mT_g} \tag{21}$$

and

$$(f-\Delta f-\nu\delta_t)(T+\Delta T) = \frac{l-\nu}{N_s} - \Delta fT\left(1+\frac{\Delta T}{T}\right) - \frac{\nu}{N_s}\frac{\Delta T}{T}. \tag{22}$$

Upon sampling, the frequency convolution integral becomes a summation with differential df equal to $\delta_r = \frac{1}{N_s(T+\Delta T)}$. The term $s^{\sqcap}(l\delta_r,\tau)$ is evaluated at the discrete frequencies l, with sample separation $\delta_r = \frac{1}{N_s(T+\Delta T)}$, and to conserve energy it must be multiplied by δ_r. Thus, sampling (4) at frequency $l=k$ gives the following result:

$$Y(k) = \frac{1}{N_s^2(T+\Delta T)}\sum_{d=-\frac{N_s}{2}}^{\frac{N_s}{2}-1}\sum_{\nu=-\frac{N_s}{2}}^{\frac{N_s}{2}-1} X_m(\nu) \cdot e^{\jmath 2\pi(d-\nu)\delta_t mT_g} \cdot e^{-\jmath 2\pi\left[\frac{d-\nu}{N_s}-\frac{\nu}{N_s}\frac{\Delta T}{T}-\Delta fT(1+\frac{\Delta T}{T})\right]\left(mN_g+\frac{N_s-1}{2}\right)} \cdot\frac{\sin\left\{\pi\left[\frac{d-\nu}{N_s}-\frac{\nu}{N_s}\frac{\Delta T}{T}-\Delta fT(1+\frac{\Delta T}{T})\right]N_s\right\}}{\sin\left\{\pi\left[\frac{d-\nu}{N_s}-\frac{\nu}{N_s}\frac{\Delta T}{T}-\Delta fT(1+\frac{\Delta T}{T})\right]\right\}} \cdot\int_0^\infty s^{\sqcap}((k-d)\delta_r,\tau)e^{-\jmath 2\pi\nu\delta_t\tau}d\tau + N(k). \tag{23}$$

If we now separate out $\nu=d$ from the rest of the terms in the second summation of (23), we obtain

$$Y(k) = \frac{1}{N_s(T+\Delta T)}\sum_{d=-\frac{N_s}{2}}^{\frac{N_s}{2}-1} X_m(d)\lambda(d,d) \cdot\int_0^\infty s^{\sqcap}((k-d)\delta_r,\tau)e^{-\jmath 2\pi d\delta_t\tau}d\tau + \frac{1}{N_s(T+\Delta T)}\sum_{d=-\frac{N_s}{2}}^{\frac{N_s}{2}-1}\sum_{\substack{\nu=-\frac{N_s}{2}\\ \nu\neq d}}^{\frac{N_s}{2}-1} X_m(\nu)\lambda(d,\nu) \cdot\int_0^\infty s^{\sqcap}((k-d)\delta_r,\tau)e^{-\jmath 2\pi\nu\delta_t\tau}d\tau + N(k), \tag{24}$$

in which $\lambda(d,\nu)$ represents the phase and magnitude effects (note that $\lambda(d,\nu)$ absorbs one of the $\frac{1}{N_s}$ factors of (23)). This function is given in (7).

Finally, by isolating the terms $d=k$ from both sums on d in (24) and by using the definition of the time-domain average channel given in (8), we obtain (6), (9), (10) and (11).

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